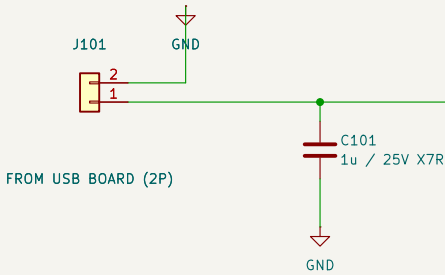


	1	2	3	4	5	6	7	8		
A	TRIMIX ANALYSER — SYSTEM OVERVIEW A2.2 engineering review / Double-click a numbered block to open its circuit page.									
	POWER FLOW USB / protected pack > charger SYS > switched 5 V > Guition and sensors									
B	01 CHARGING + BATTERY BQ25895 / protected 1S2P holder 2 x 3400 mAh = 6800 mAh nominal RCY/BEC pigtail + cell thermistor		02 SWITCHED 5 V TPS63020 buck-boost Powers Guition and gas sensors Battery assists when USB power is limited		03 GUITION + FUEL GAUGE MAX17048 + JP1 host harness Separate 100 kHz sensor I2C bus UART + shutdown / enable signals					
C	04 TWO OXYGEN INPUTS A02 cable + insulated SMB coax One ADS1115, two differential channels Independent calibration; sequential reads		05 HELIUM BRIDGE MD62 thermal-conductivity bridge Regulated 3.0 V excitation + ADS1115 Known-gas calibration required		06 GAS-CHAMBER CLIMATE Wired GYBMEP / BME280 module Temperature, pressure and humidity Verify BME chip ID and module voltage					
D	07 EXPERIMENTAL CO Wired ZE07-CO + translated UART Dedicated supply with voltage margin Not a breathing-gas safety clearance		08 PUSH-BUTTON ON / OFF LTC2954-1 + momentary 1NO button Save state, then switch off the 5 V rail Held button provides forced shutdown		09 USB-A + USB-C INPUT GCT USB4720 + two CC resistors Separate 0.60 mm USB daughterboard 5 V input from USB-A or USB-C					
E	COMMISSIONING Charge ARM stays open until cell / holder / USB checks pass. Guition USB must be disconnected for full shutdown.									
F	Native schematic + unrouted 3D placement preview. Connector dimensions, thermal design and gas accuracy require bench validation.						Engineering review draft — not released for manufacture Trimix Sheet: / File: Trimix_Analyzer.kicad_sch Title: Trimix analyser — system overview Size: A3Date: 2026-09-06Rev: A2.2 KiCad E.D.A. 10.0.6Id: 1/10			
	1	2	3	4	5	6	7	8		

Charging & protected battery

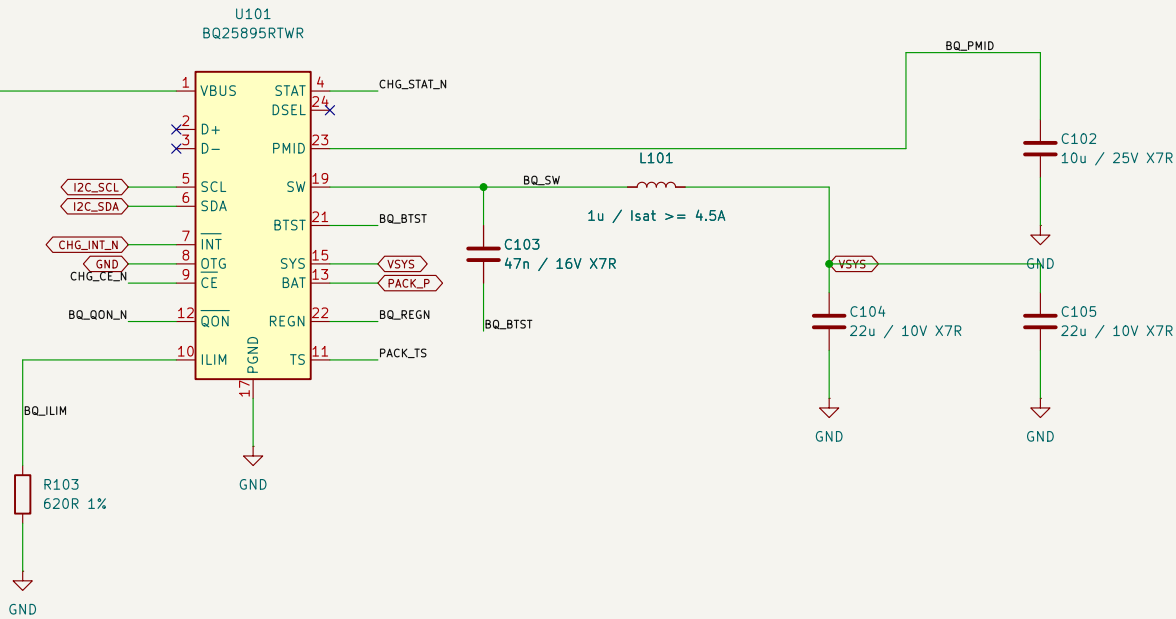
Follow the top row from USB input to charger and battery. The lower row controls and monitors charging.

01 USB INPUT

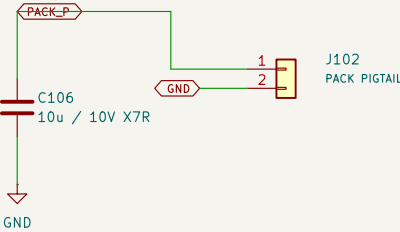


J101 receives 5 V from the GCT USB daughterboard.
J902 -> two-wire harness -> J101 (same pin numbers).
See USB_input, sheet 10; qualify before charging.

02 CHARGER & POWER PATH

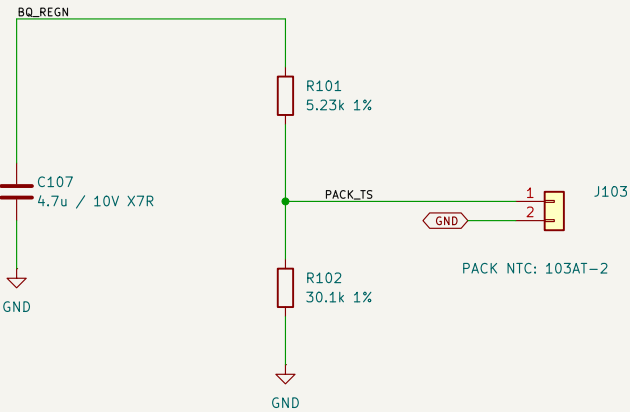


03 PROTECTED BATTERY



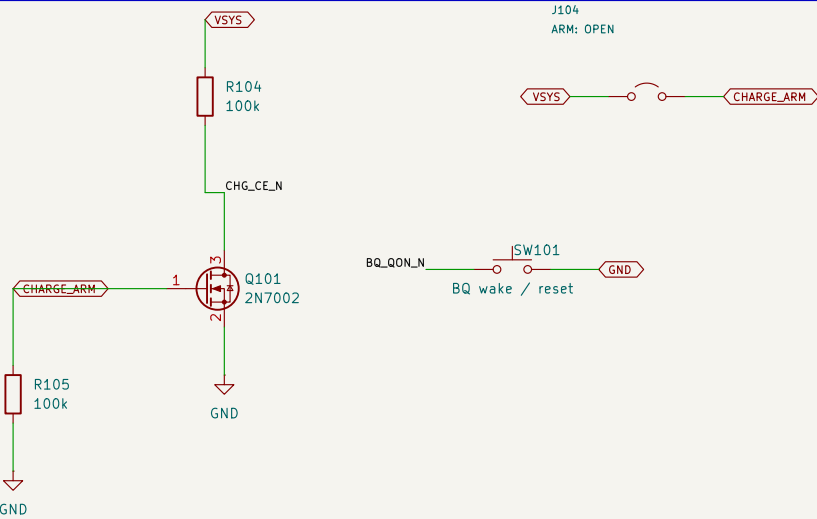
J102 = PCB pigtail to the RCY/BEC cable mate.
Red = protected P+; black = protected P-.
FMA holder: 2 x 3400 mAh in parallel, 6800 mAh total.
Verify polarity; no PCB-mounted RCY footprint is implied.

04 PACK TEMPERATURE



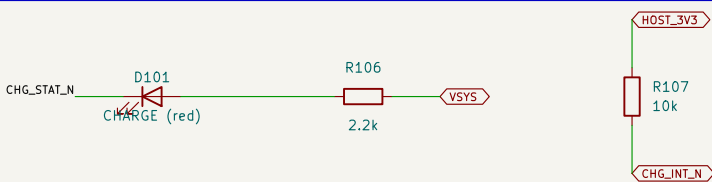
R101/R102 and the external 10k NTC sense temperature.
Attach the NTC to the cells; connect it at J103.
An open sensor inhibits charging.

05 COMMISSIONING ARM & SERVICE WAKE



J104 OPEN = charging disabled; R105 holds Q101 off.
Fit the shunt only after cell/USB/NTC validation.
Armed = autonomous charging with the host ON or OFF.
SW101 is service wake/reset, not the front power button.

06 STATUS INDICATORS



D101 lights while the charger reports charging.
R106 limits LED current.
R107 pulls the interrupt up to the host 3.3 V rail.

Autonomous defaults: 4.208 V, 2.048 A charge setting.
Floating D+/D- selects 500 mA input; actual charge
current falls to match input power. ILIM: 620R (~0.57 A).
No charger firmware is implemented; defaults need review.

READING THIS SHEET

Matching labels connect electrically. VSYS also supplies the always-on pushbutton controller.
BQ25895 I2C address = 0x6A. If firmware writes settings, disable its watchdog before host shutdown.
Default settings return after complete charger power loss. Keep J104 OPEN until those settings are qualified.

Engineering review draft | Charge-arm shunt initially OPEN

Trimix

Sheet: /01 CHARGING + BATTERY/
File: Charging.kicad_sch

Title: USB charging and protected 1S2P pack

Size: A3 Date: 2026-09-06

KiCad E.D.A. 10.0.6

Rev: A2.2

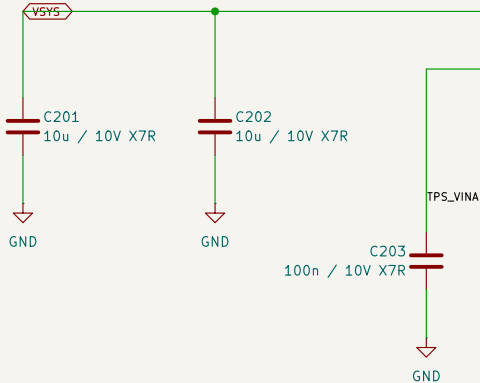
Id: 2/10

Regulated 5 V supply

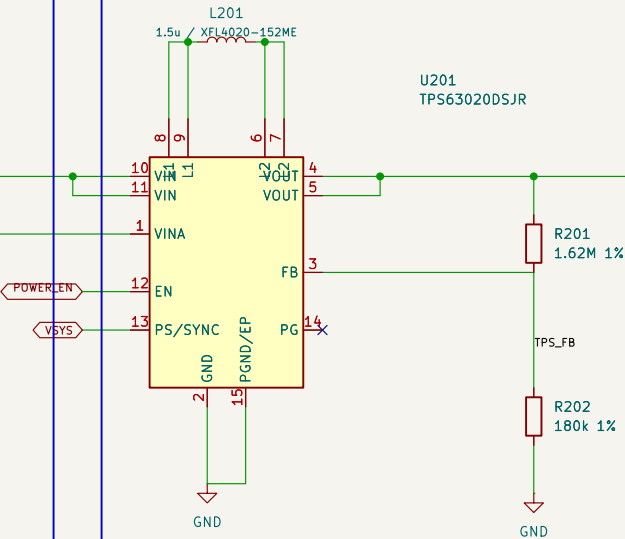
VSYS from sheet 2 is converted to VOUT_5V for the Guition board. The target is 5 V / 1 A, subject to validation.

01 INPUT FILTERING

C201/C202 provide local input energy.
C203 bypasses the internal control supply.

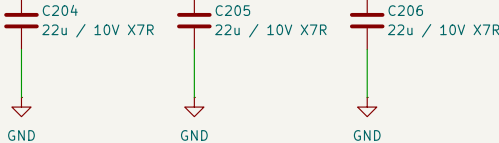


02 CONVERTER & FEEDBACK

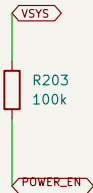


03 OUTPUT FILTERING

C204-C206 support load steps
and reduce output ripple.



04 PUSHBUTTON CONTROLLER ENABLE



POWER_EN comes from the LTC2954-1 on sheet 9.
R203 pulls EN up; the controller pulls it low to turn off.
Only the regulated 5 V output is switched.

05 HOW THE VOLTAGE IS SET

$5,000\text{ V} = 0.5\text{ V} \times (1 + 1.62\text{M} / 180\text{k})$
R201/R202 feed a fraction of the output back to FB.
Use 1% resistors; confirm the output on the bench.

Engineering review draft | Charge-arm shunt initially OPEN

Trimix

Sheet: /02 SWITCHED 5 V/

File: Supply_5V.kicad_sch

Title: 5 V buck-boost supply

Size: A4 Date: 2026-09-05

KiCad E.D.A. 10.0.6

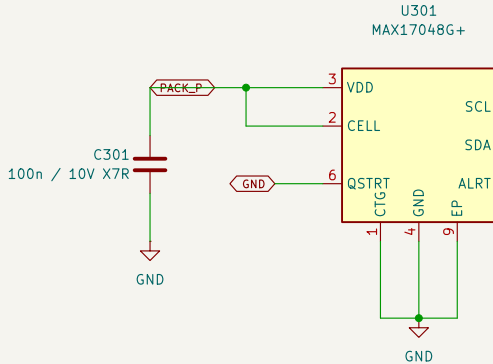
Rev: A2

Id: 3/10

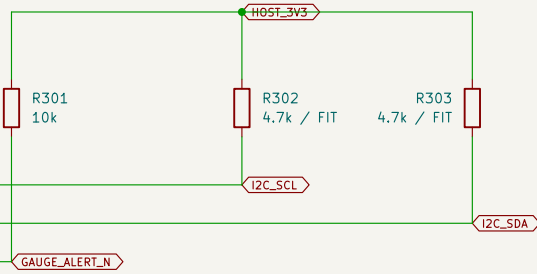
Battery gauge & host interface

The fuel gauge reads the protected pack. All communication pull-ups use the Guition board's 3.3 V supply.

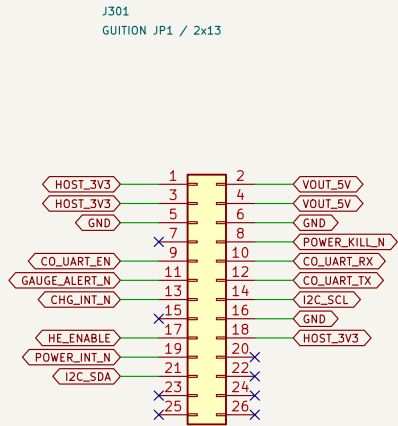
01 BATTERY GAUGE



02 I2C & ALERT PULL-UPS



03 GUITION JP1 – POWER & SIGNALS



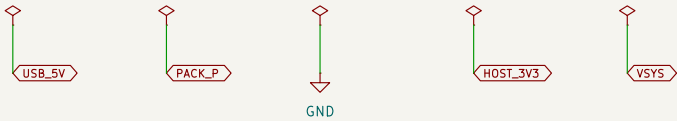
Pin numbers match the vendor JP1 drawing. Confirm the physical mating view.
5 V flows TO Guition; HOST_3V3 returns FROM its switched 3.3 V regulator.

GPIO28 / pin21 = I2C SDA GPIO29 / pin14 = I2C SCL
GPIO30 / pin12 = CO TX GPIO31 / pin10 = CO RX (host perspective)
GPIO32 / pin19 = power INT GPIO33 / pin8 = power KILL
GPIO34 / pin17 = He enable GPIO49 / pin13 = charger INT
GPIO50 / pin11 = gauge ALT GPIO51 / pin9 = CO UART enable

JP1 pins 23/25 belong to the existing GPIO7/8 touch bus and are unused here.
GPIO35 (BOOT), GPIO52 and ESP32–C6 pins are deliberately unconnected.
The firmware must add the new GPIO28/29 I2C bus; it is not implemented.

Disconnect Guition USB power for the button to switch the device fully off.
Do not connect the protected pack to Guition CN4 / its onboard charger.
Connector pitch, mechanical mate and cable ampacity remain to be verified.

04 ERC POWER SOURCE MARKERS



PACK_P = protected red P+ / GND = protected black P–.
Two 3400 mAh parallel cells: 6800 mAh nominal, one voltage stack.

Engineering review draft | Charge–arm shunt initially OPEN

Trimix

Sheet: /03 GUITION + FUEL GAUGE/
File: Gauge_Interface.kicad_sch

Title: Battery gauge and Guition interface

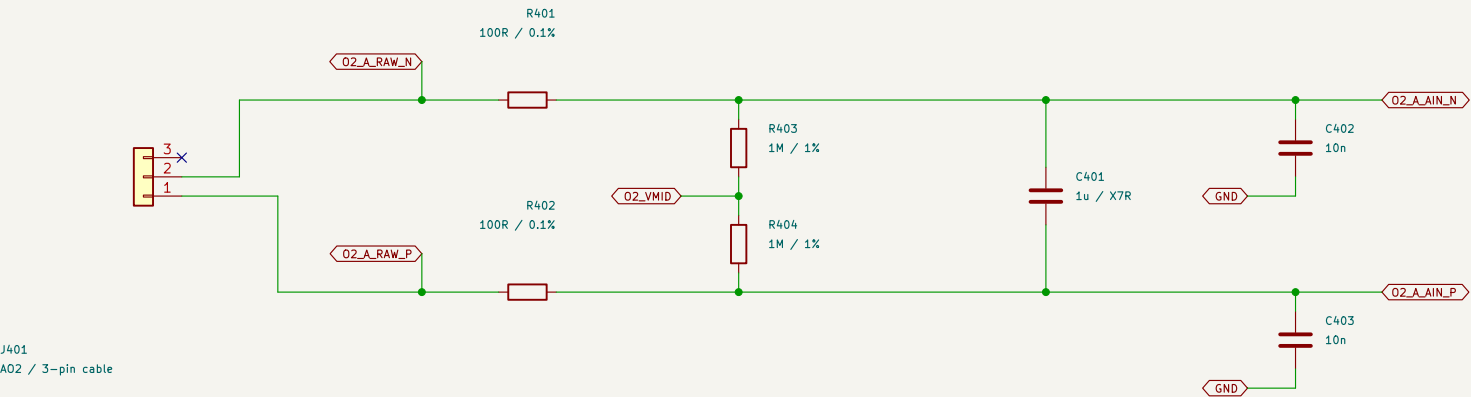
Size: A3 Date: 2026–09–05
KiCad E.D.A. 10.0.6

Rev: A2
Id: 4/10

OXYGEN | TWO INDEPENDENT MILLIVOLT INPUTS

Passive galvanic cells in a vented chamber – short cables <=30 cm – review draft

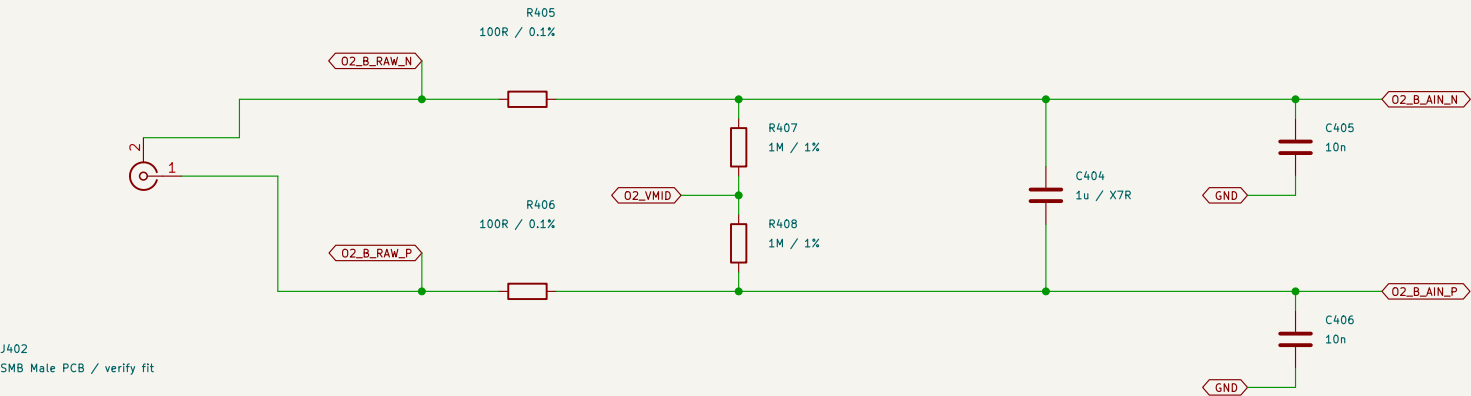
1 | OXYGEN A – AO2 CABLE



J401
AO2 / 3-pin cable

PCB harness convention: 1 = SENSE+, 2 = SENSE-, 3 = spare/NC. Verify all AO2 cable pins before mating.

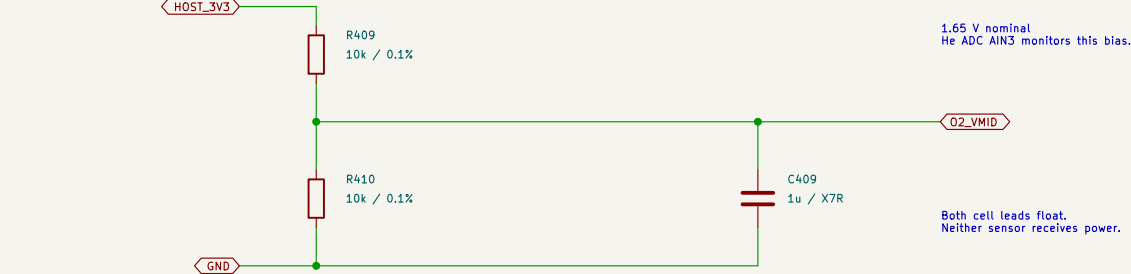
2 | OXYGEN B – ISOLATED COAX



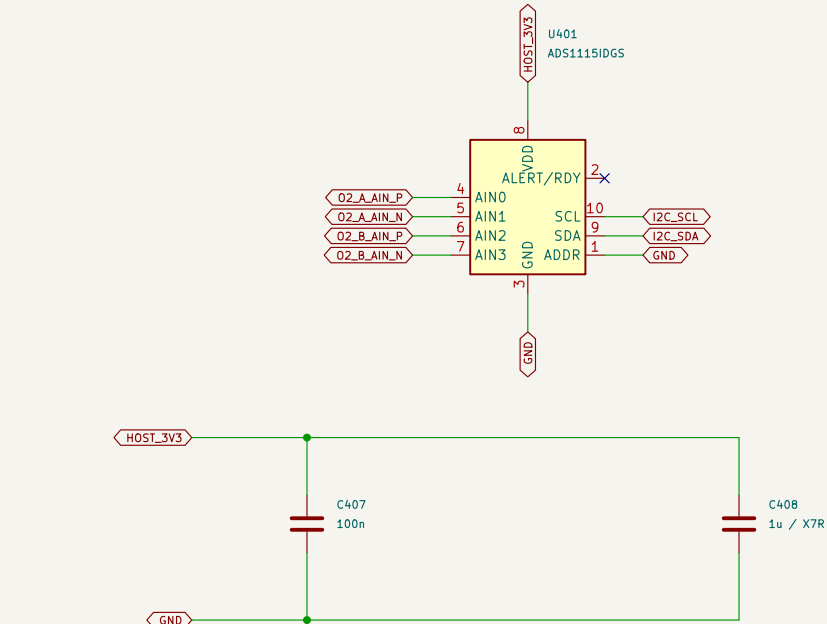
J402
SMB Male PCB / verify fit

SMB shell = SENSE-, Insulate it from the enclosure and GND; no 50 ohm load.

4 | SHARED MID-SUPPLY BIAS



3 | DIFFERENTIAL ADC – I2C 0x48



Place C407 at U401 VDD/GND. I2C pull-ups are on the host sheet.

5 | SETUP AND CABLE RULES

AIN0-AIN1 = O2 A; AIN2-AIN3 = O2 B.
Start at +/-0.256 V and 8 SPS; keep signed readings.
One ADC multiplexes the pairs; they are not simultaneous.

No 50 ohm termination. No ground connection on SMB shell.
Check cell loading, reversed leads and power-off leakage
with the actual sensors before accepting readings.

J402 is the owned SMB Male PCB connector. Verify its
land pattern and mating sensor plug before manufacture.
A disconnected/shorted cell can resemble 0% O2;
calibration and plausibility checks are required.

Engineering review draft – not released for manufacture

Trimix

Sheet: /04 TWO OXYGEN INPUTS/
File: Oxygen.kicad_sch

Title: Oxygen | two independent millivolt inputs

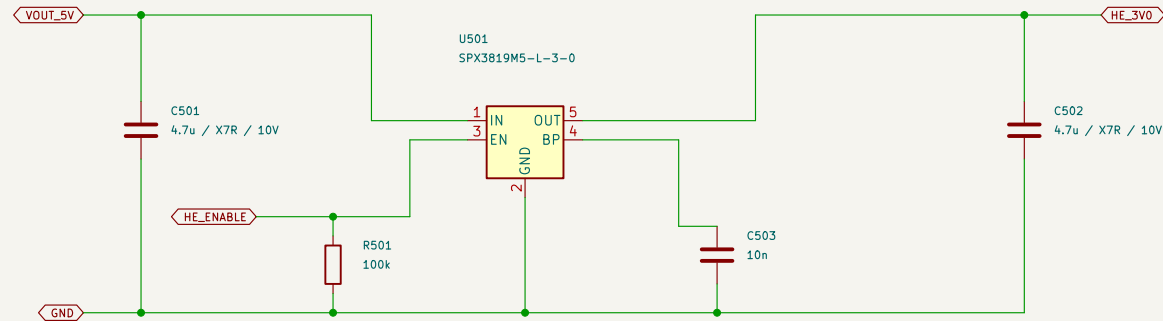
Size: A3 Date: 2026-09-05
KiCad E.D.A. 10.0.6

Rev: A2
Id: 5/10

HELIUM | MD62 THERMAL-CONDUCTIVITY BRIDGE

Winsen MD62 repurposed for He – constant 3.0 V excitation – calibration-dependent review draft

1 | SWITCHED 3.0 V EXCITATION

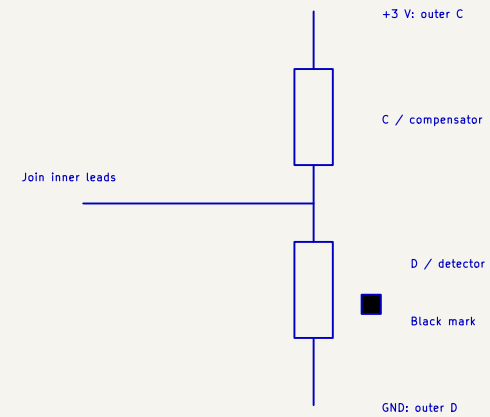


3.0 V $\pm$ 0.1 V at the sensor; operating current $\leq$ 120 mA.
Enable after HOST_3V3 is valid; disable before the ADC loses power.
LDO may dissipate 0.24 W. Verify temperature and cable voltage drop.

2 | REMOTE MD62 – THREE-WIRE HARNESS

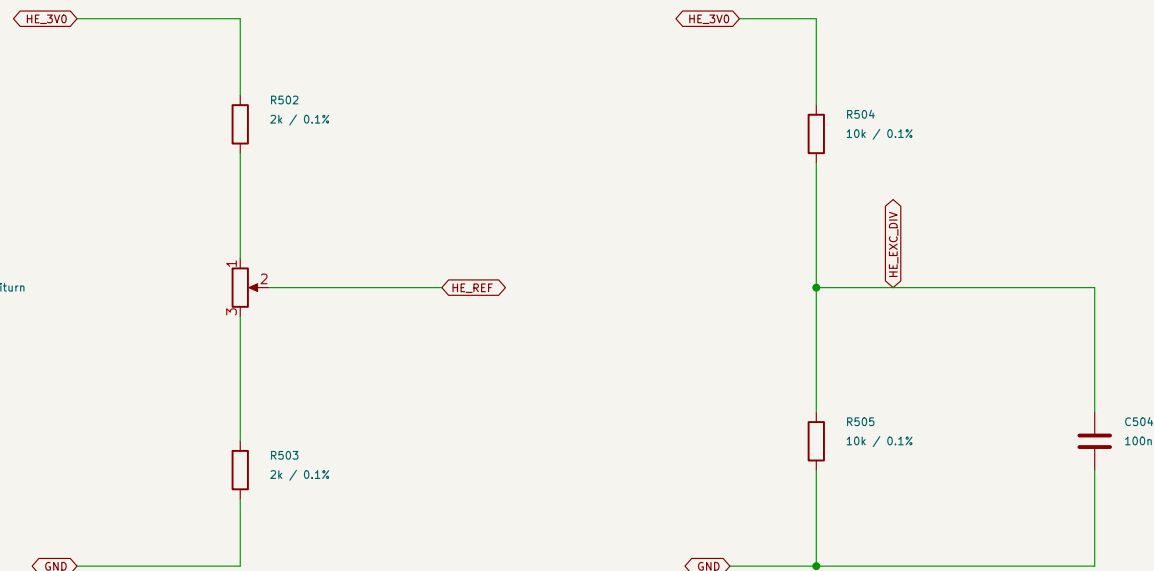


PCB assignment:
1 = +3 V, 2 = joined midpoint, 3 = GND.
MD62 has no assumed numeric lead order.



Winsen Manual V1.3: bridge + marked detector lead, pp. 1 and 3.

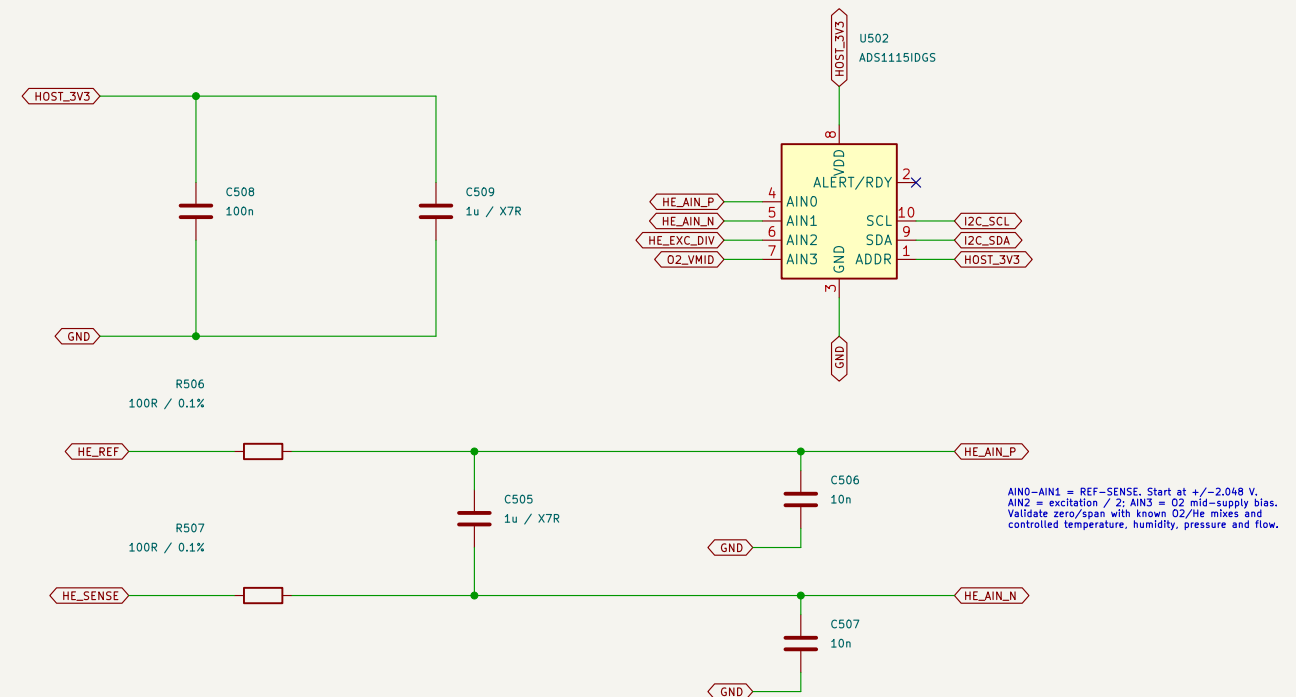
3 | BRIDGE REFERENCE AND EXCITATION CHECK



Reference branch: 2k + 500R trim + 2k across the same 3 V.
Adjust near air balance; record signed offset and span during calibration.
HE_EXC_DIV = board excitation / 2; it does not sense remote cable drop.

MD62 is a thermal-conductivity sensor, not selective to He. Conditioning and gas-specific calibration are mandatory; no factory He accuracy is specified.

4 | DIFFERENTIAL ADC – I2C 0x49



Engineering review draft – not released for manufacture

Trimix

Sheet: /05 HELIUM BRIDGE/

File: Helium.kicad_sch

Title: Helium | MD62 thermal-conductivity bridge

Size: A3

Date: 2026-09-05

Rev: A2

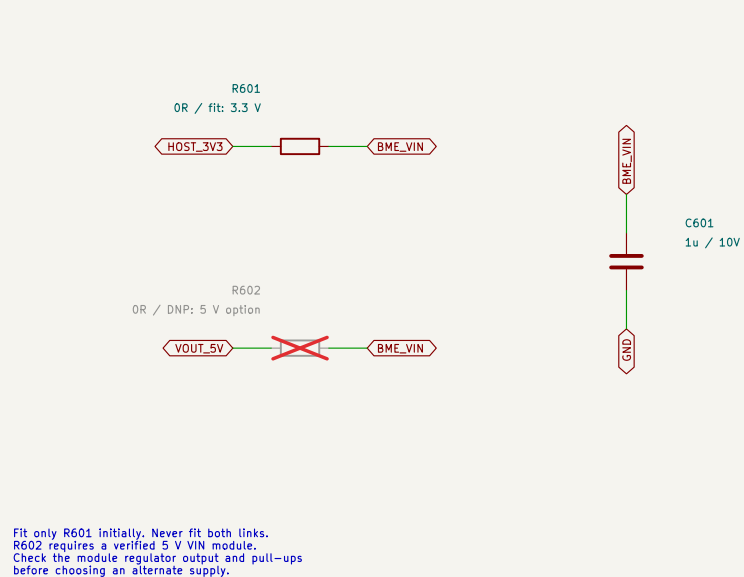
KiCad E.D.A. 10.0.6

id: 6/10

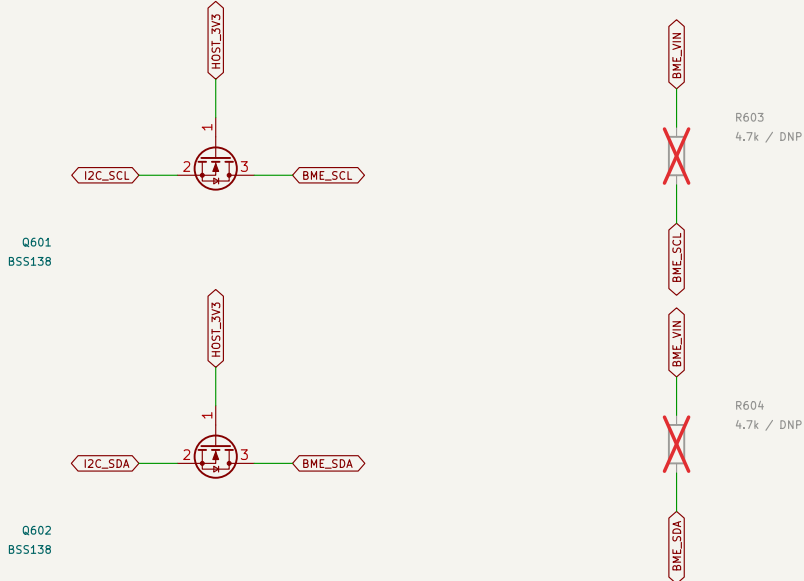
WIRED HUMIDITY, TEMPERATURE & PRESSURE

GYBMEP module in the vented gas chamber. Four-wire harness, maximum 30 cm; dedicated sensor I2C at 100 kHz.

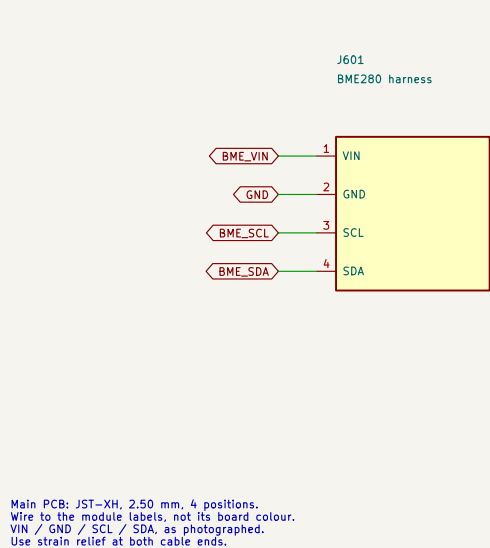
01 MODULE POWER SELECTION



02 I2C LEVEL INTERFACE



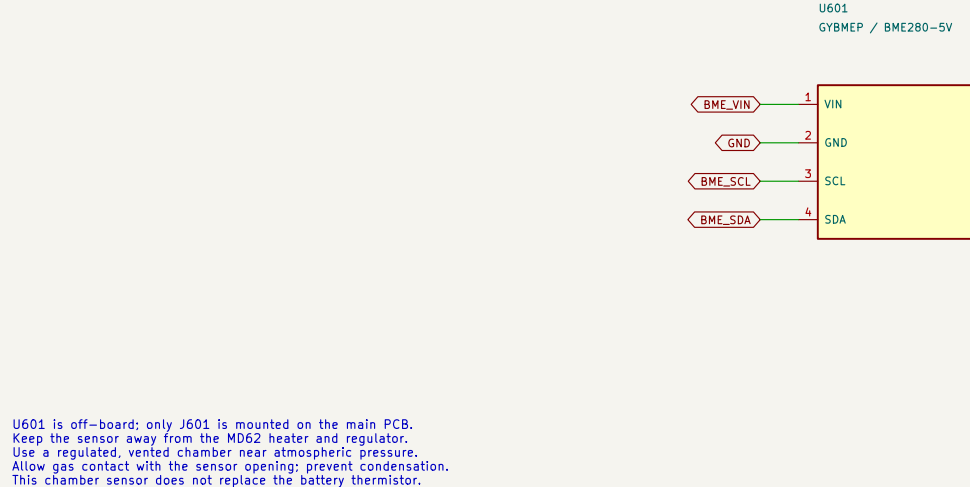
03 PCB HARNESS CONNECTION



04 COMMISSIONING & SIGNAL INTEGRITY

1. Read chip ID register 0xD0; BME280 must return 0x60. BMP280 does not measure humidity. Probe 0x76 and 0x77.
2. Verify the actual module VDD / VDDIO and pull-up rail. Chip limits; VDD 1.71–3.6 V; VDDIO 1.2–3.6 V. The product photograph does not prove the fitted regulator.
3. R603/R604 are unpopulated initially; use module pull-ups. Fit only after checking their rail and the installed resistance. Host pull-ups are on the host interface sheet.
4. Measure rise time with the complete cable: <=1 us at 100 kHz. Keep total bus capacitance within the device / I2C limits. BSS138 translates logic; it does not isolate a stuck-low bus.

05 REMOTE MODULE IN THE CHAMBER



Engineering review draft – not released for manufacture

Trimix

Sheet: /06 GAS-CHAMBER CLIMATE/
File: Environment.kicad_sch

Title: Wired humidity, temperature & pressure

Size: A3 Date: 2026-09-05
KiCad E.D.A. 10.0.6

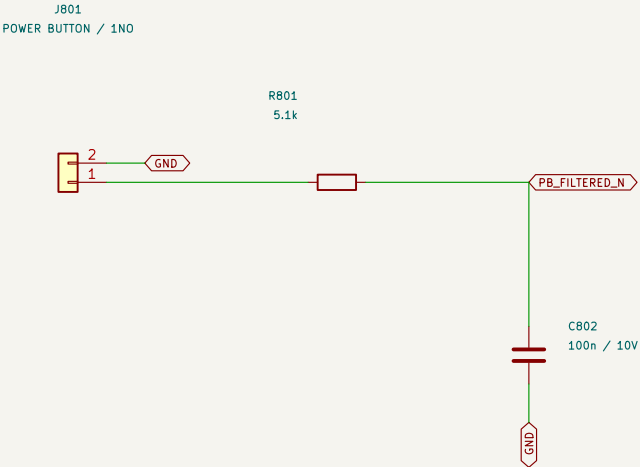
Rev: A2
Id: 7/10

id: 8/10

MOMENTARY POWER BUTTON & SAFE SHUTDOWN

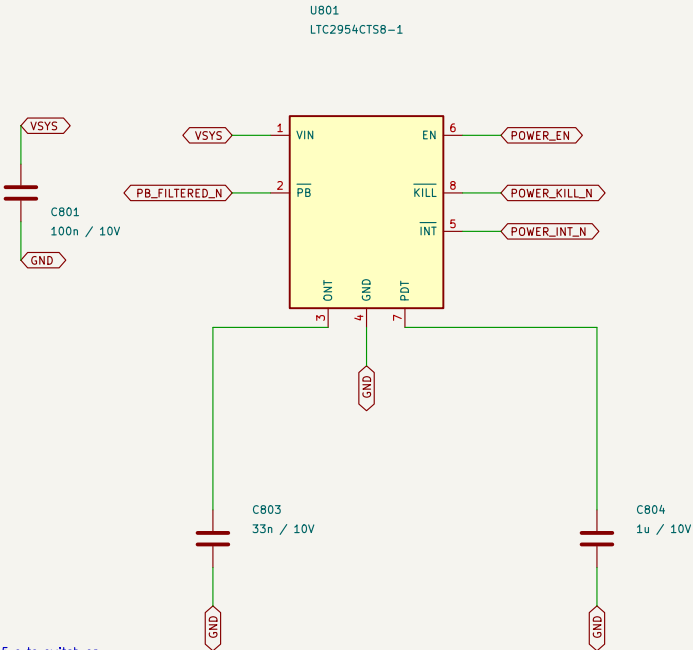
The button switches the 5 V load. The armed charger remains available with the screen and sensors off.

01 PANEL BUTTON & INPUT FILTER



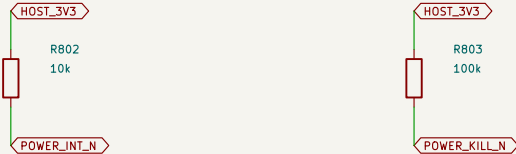
J801 carries only the isolated switch contacts. The separate LED wires go to J802. Place R801/C802 close to U801. A normally-open contact closes PB to ground.

02 ALWAYS-ON POWER LATCH



ONT: about 0.245 s to switch on.
PDI: about 6.5 s held to force off.
R203 on sheet 3 supplies the EN pull-up.

03 HOST SHUTDOWN HANDSHAKE



GPIO32 reads the debounced active-low interrupt. GPIO33 releases KILL during boot and normal use. After saving state, drive KILL low to turn off. Configure GPIO33 as open-drain; avoid a low glitch. R803 uses the switched 3.3 V rail; so startup does not depend on firmware asserting an early hold.

HOST_3V3 must rise within the 400 ms minimum startup blanking period; verify on real hardware.

04 BUTTON SEQUENCE & IMPLEMENTATION

OFF: press for about 0.25 s; the converter, Guiton and sensors turn on.
ON: firmware should treat a deliberate press as a shutdown request, stop sensing, save state, then pull POWER_KILL_N low. This firmware is not implemented yet.
A continuous ~6.5 s hold forces off even if the host hangs. Timing has component tolerance.
The charger keeps working while OFF only after the commissioning ARM shunt is fitted.
Guiton USB must be disconnected for full power-off; do not use its CN4 battery input.

05 SEPARATE BUTTON LED



Engineering review draft – not released for manufacture

Trimix

Sheet: /08 PUSH-BUTTON ON / OFF/
File: Power_Control.kicad_sch

Title: Momentary power button & safe shutdown

Size: A3 Date: 2026-09-05
KiCad E.D.A. 10.0.6

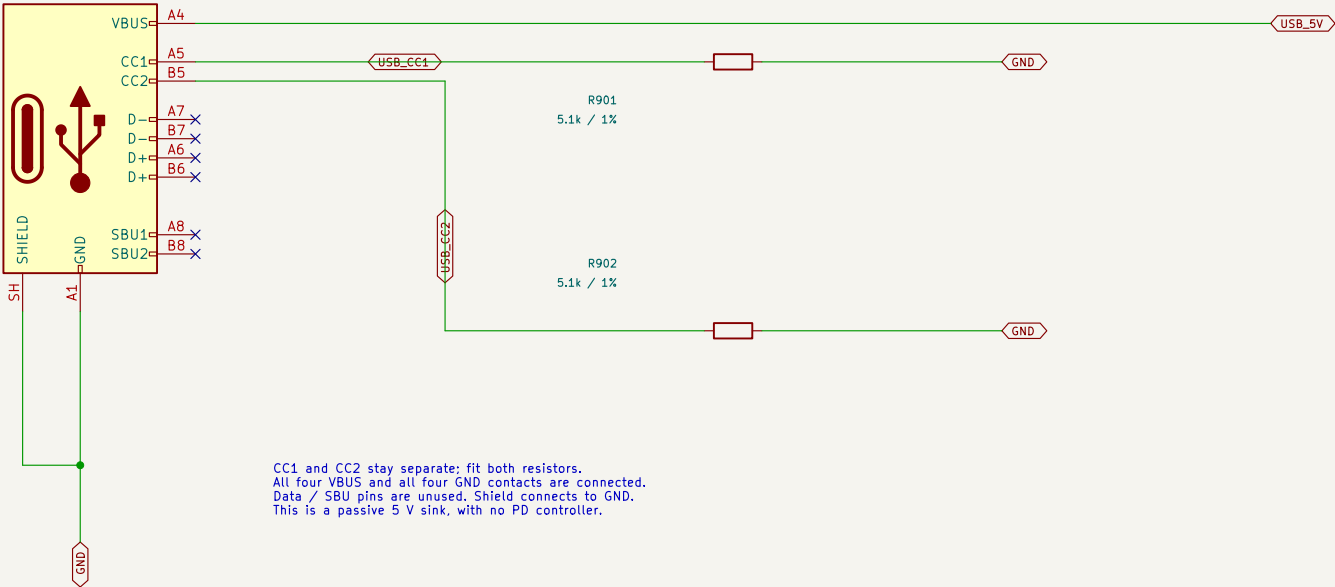
Rev: A2
Id: 9/10

USB INPUT | GCT USB4720-03-A DAUGHTERBOARD

5 V charging from USB-A or USB-C. Separate 0.60 mm USB PCB; main charger PCB remains 1.60 mm.

01 GCT SOCKET + TWO CC RESISTORS

J901
GCT USB4720-03-A



02 TWO-WIRE OUTPUT TO CHARGER

J902
TO MAIN PCB J101



J902 pin 1 -> positive wire -> J101 pin 1
J902 pin 2 -> return wire -> J101 pin 2

J901 / R901 / R902 / J902 belong to the USB PCB.
The main charger retains its two-wire input.
Use soldered leads with enclosure strain relief;
wire size and final mounting remain to be checked.

03 CABLES, ORIENTATION & CURRENT

USB-C charger + C-to-C cable: separate Rd on CC1 / CC2
allows source attachment in either plug orientation.
Test both ends flipped, including an electronically marked cable.

USB-A charging adapter + A-to-C cable uses the same input.
The compliant legacy cable supplies Rp; do not add Rp here.

BQ25895 D+/D- remain unused; nominal 500 mA INPUT setting.
Battery charge current depends on available power and system load.
No 9/12/20 V PD request or 1.5/3 A detection is implemented.

04 MECHANICS & VERIFICATION

GCT drawing: 0.60 +/- 0.10 mm PCB and edge cutout.
Do not fit USB4720 directly to the 1.60 mm main board.
Provide rigid support against cable insertion forces.
IP67 connector does not certify the enclosure; charge dry.

Before power: verify CC1-GND and CC2-GND -5.1k.
With ARM OPEN: verify 5 V in all cable orientations,
including no battery / host off. Then qualify loaded
voltage, input current and inrush. Battery charging
tests follow cell / holder / thermistor qualification.

Engineering review draft – not released for manufacture

Trimix

Sheet: /09 USB-A + USB-C INPUT/
File: USB_Input.kicad_sch

Title: USB input | GCT USB4720-03-A daughterboard

Size: A3	Date: 2026-09-06	Rev: A2.2
KiCad E.D.A. 10.0.6		Id: 10/10